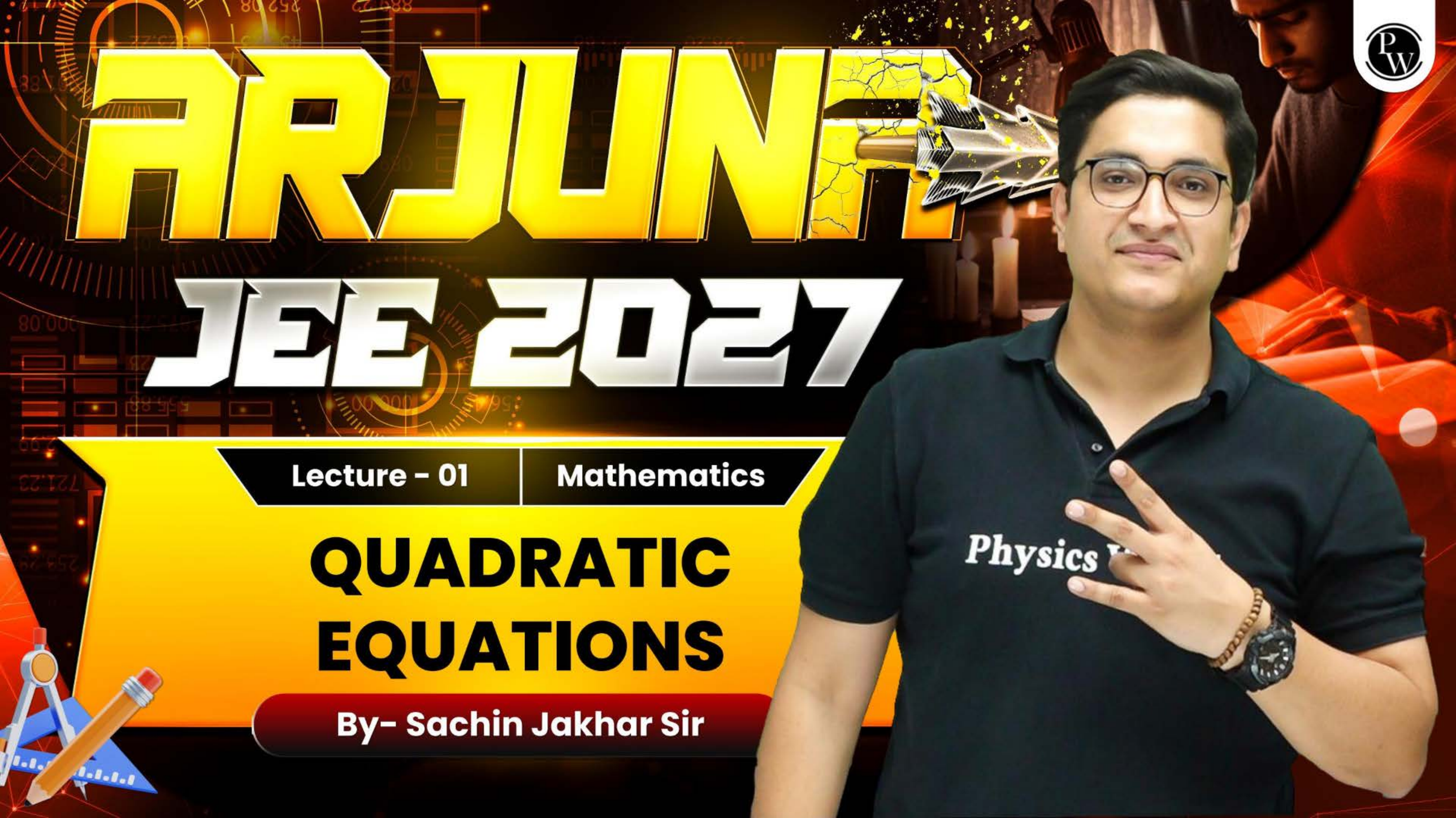
ARJUNA JEE 2027

Lecture - 01 | Mathematics

QUADRATIC EQUATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

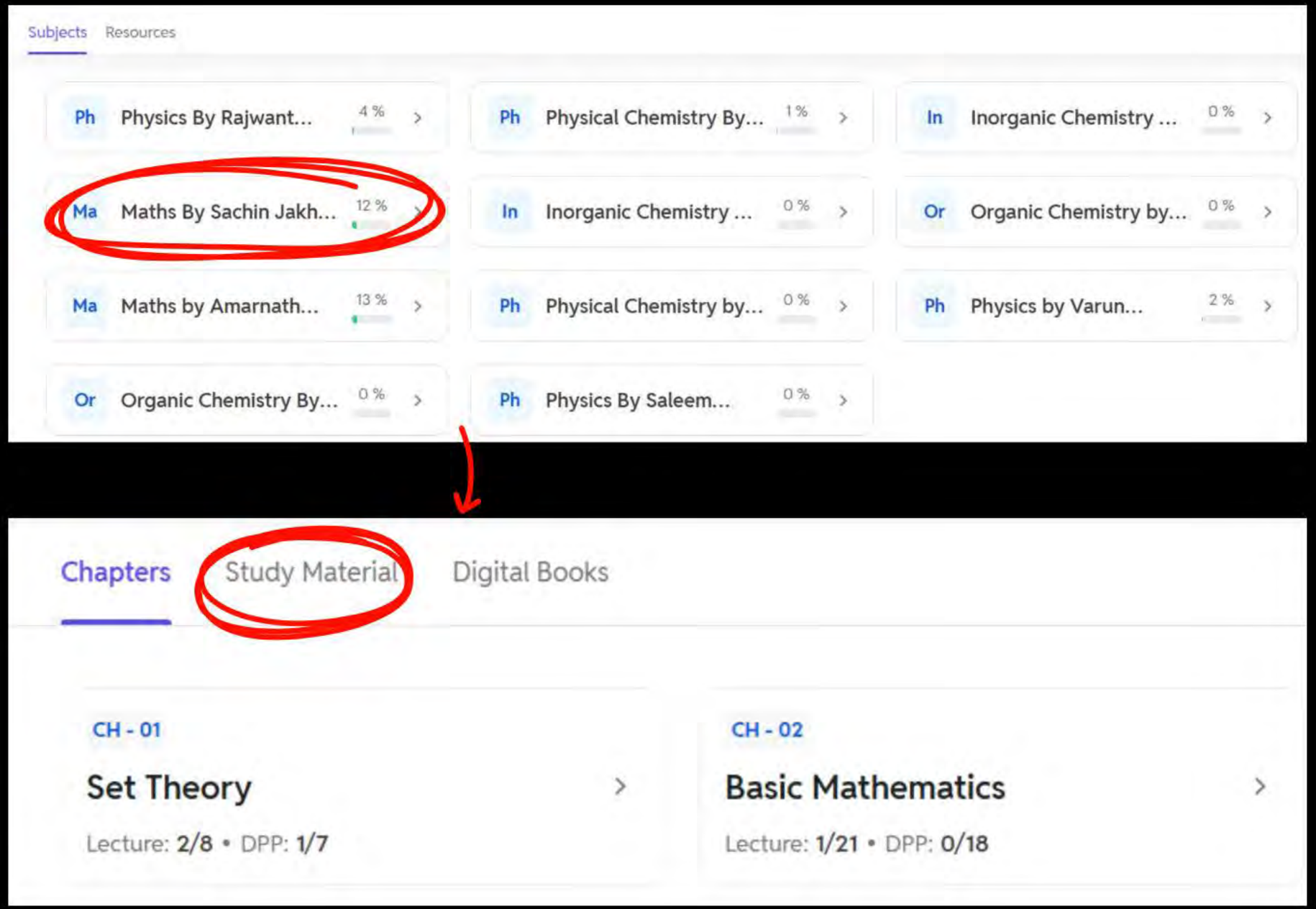
A **Irrational Inequalities**

B **Ratio's**

C **Introduction to Quadratic Equation**

D **Question Practice**




Subjects
Resources
Ph Physics By Rajwant... 4 %
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SM - 01
PYQ Practice Sheet || Only PDF
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SM - 07
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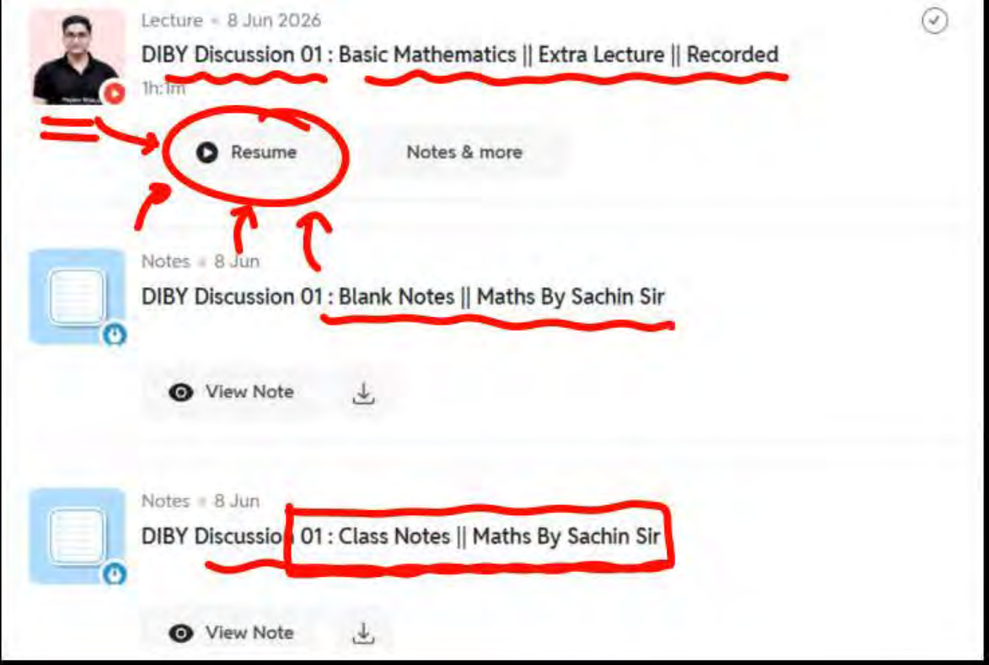
SM - 08
PYQ Practice Sheet
2 DPP • 2 Notes

SM - 09
Mission 11th Comeback
1 Lectures • 1 Notes

SM - 10
DIBY Discussion By Sachin Sir
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SM - 11
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2 Notes

SM - 12
JEE Replica Sheet
2 DPP • 2 Notes





LAST CLASS RECAP

Answer the following Questions:

1. Square Root is only defined for non-negative numbers or expressions. → (T)
2. Squaring may leads to extra solutions, so always check your answer. ↳ (T)
3. Value of square root is non-negative. → (T)
4. Squaring in an inequality is only allowed when both sides are non-negative
5. Values of x for which $f(x) = \sqrt{x-2} + \sqrt{5-x}$ is defined are $x \in [2,5]$ ✓.

$x-2 \geq 0$ ↑ $5-x \geq 0$

$x \geq 2$ | $5 \geq x$

QUESTION PRACTICE

Acidic Question

$x = 6 \rightarrow$ LHS $= 2$, RHS $= 2$ ✓

$x = \frac{22}{9}$ ✓ LHS $= \frac{4}{3}$, # RHS $= \frac{4}{3}$

Solve following equation:

(ii) $\sqrt{x+3} - 1 = \sqrt{x - \sqrt{x-2}}$

SBS

$(x+3) + 1 - 2\sqrt{x+3} = x - \sqrt{x-2}$

$4 - 2\sqrt{x+3} = -\sqrt{x-2}$

SBS

$16 + 4(x+3) - 16\sqrt{x+3} = x - 2$

$4x + 28 - 16\sqrt{x+3} = x - 2$

$3x + 30 = 16\sqrt{x+3}$

SBS

$9x^2 + 900 + 180x = 256x + 256 \times 3$

$9x^2 - 76x + 44 \times 3 = 0$

$9x^2 - 54x - 22x + 44 \times 3 = 0$

$9x(x-6) - 22(x-6) = 0$

$(x-6)(9x-22) = 0$

$x = 6, \frac{22}{9}$

Question

Complete solution set of Inequality : $2 - \frac{2x}{3} \geq \sqrt{x+4}$

WCM

M-2

domain

$x+4 \geq 0$

$x \geq -4$

A $x \in (-\infty, 0]$

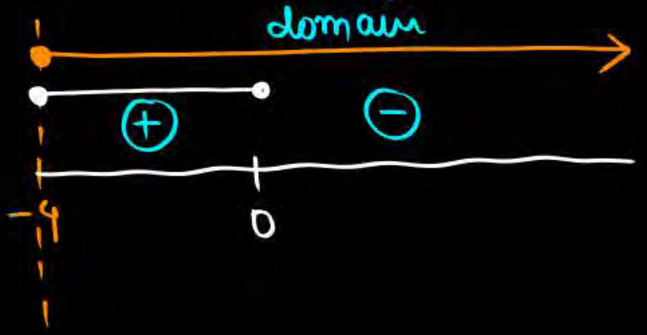
B $x \in [-4, 0) \cup \left(\frac{33}{4}, \infty\right)$

C $x \in [-4, 0]$

D $x \in [0, \infty)$

eq^n → Roots

$2 - \frac{2x}{3} - \sqrt{x+4} \geq 0$



$2 - \frac{2x}{3} = \sqrt{x+4}$

SBS

$4 + \frac{4x^2}{9} - \frac{8x}{3} = x + 4$

$\frac{4x^2}{9} - \frac{11x}{3} = 0$

$\frac{4x^2 - 33x}{9} = 0$

$x = 0, \frac{33}{4}$

✓ ✗

Question

Complete solution set of Inequality : $\frac{4x-5}{3} > \sqrt{3-x}$

A $x \in (-\infty, 3]$

B $x \in (2, 3)$

C $x \in (2, 3]$

D $x \in \left(-\infty, -\frac{1}{16}\right) \cup (2, 3]$

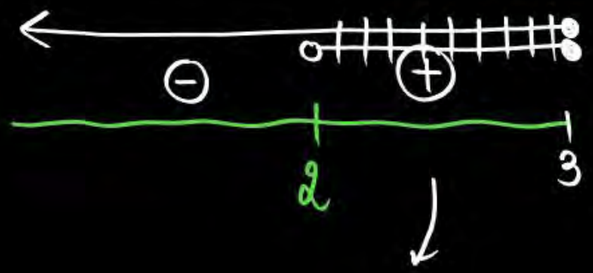
domain

$3 - x \geq 0$

$3 \geq x$

WCM :

$\frac{4x-5}{3} - \sqrt{3-x} > 0$



$x \in (2, 3]$

Roots

$\frac{4x-5}{3} = \sqrt{3-x}$

SBS

$16x^2 + 25 - 40x = 9(3-x)$

$= 27 - 9x$

$16x^2 - 31x - 2 = 0$

$16x^2 - 32x + x - 2 = 0$

$16x(x-2) + 1(x-2) = 0$

$(16x+1)(x-2) = 0$

$x = 2, -\frac{1}{16}$

• RATIO

Given: $a : b :: c : d$

OR

$\frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$

$\frac{ax}{bx} = \frac{cy}{dy} = \frac{ax+cy}{bx+dy}$

ex:- $\frac{1}{2} = \frac{7}{14} = \frac{11}{22}$

$\frac{1}{2} = \frac{7}{14} = \frac{11}{22} = \frac{1+7+11}{2+14+22} = \frac{19}{38}$

$\frac{x}{2x} = \frac{7y}{14y} = \frac{11z}{22z} = \frac{x+7y+11z}{2x+14y+22z}$ ✓

• RATIO

Given: $a:b :: c:d$

OR

$\# \quad \frac{a}{b} = \frac{c}{d}$

$+1$

$\frac{a}{b} + 1 = \frac{c}{d} + 1 \longrightarrow \frac{a+b}{b} = \frac{c+d}{d}$

-1

$\frac{a}{b} - 1 = \frac{c}{d} - 1 \longrightarrow \frac{a-b}{b} = \frac{c-d}{d}$

$\frac{a}{b}$

Reciprocal

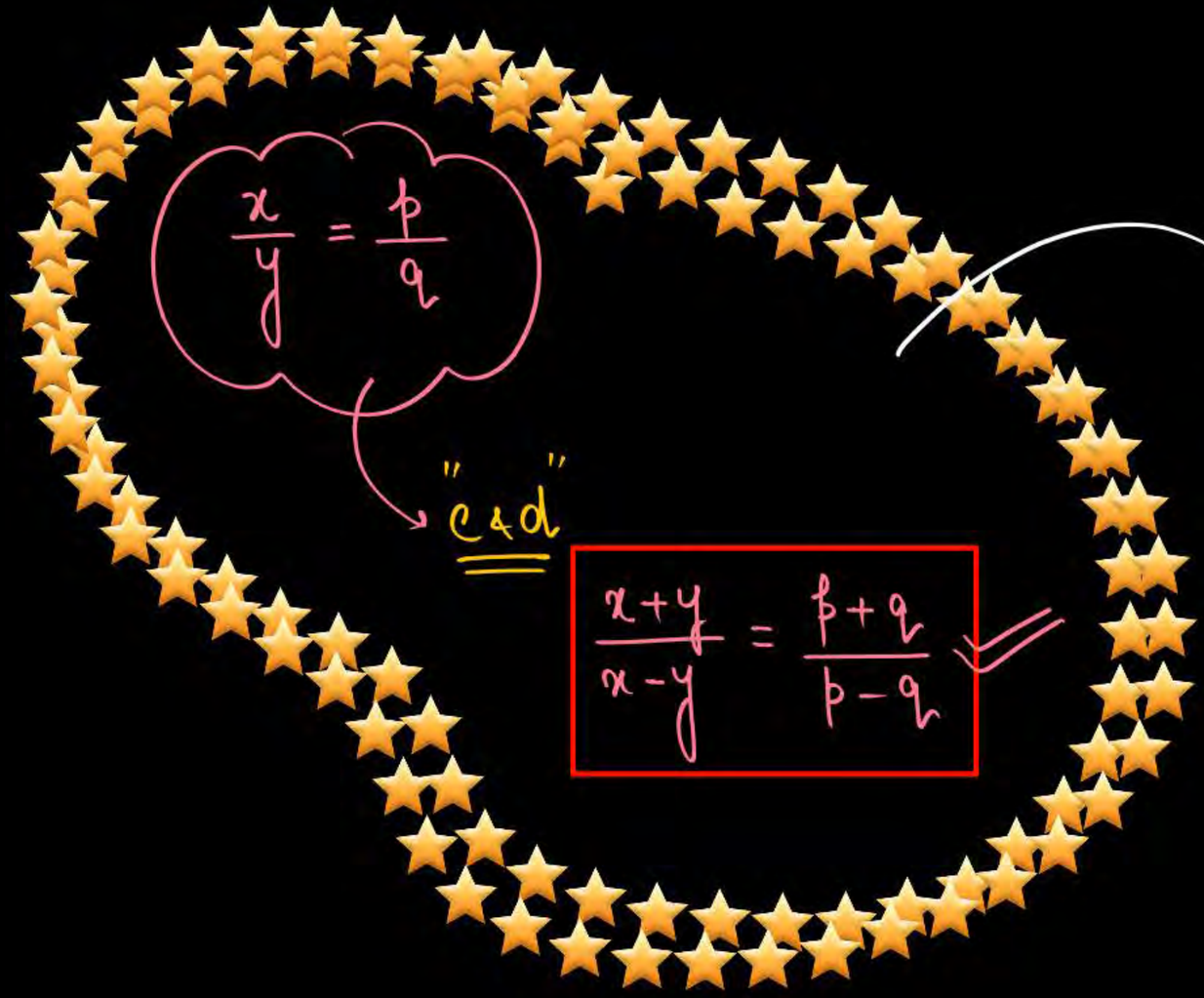
$\frac{b}{a} = \frac{d}{c}$

Alternando

Comp. & Divid.

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

$$\frac{a+b}{\cancel{b}\left(\frac{a-b}{\cancel{b}}\right)} = \frac{c+d}{\cancel{d}\left(\frac{c-d}{\cancel{d}}\right)}$$



Hint:

When $N^r + D^r$ have common Terms



RATIOS

Note: 01 If $a : b = c : d \Rightarrow \frac{a}{b} = \frac{c}{d}$

Note: 02 Componendo & Dividendo (C & D)

Note: 03 If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} \ldots\ldots\ldots$, then $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \ldots\ldots = \frac{a+c+e+\ldots.}{b+d+f+\ldots.}$

Note: 04 If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} \ldots\ldots\ldots$, then $\frac{xa}{xb} = \frac{yc}{yd} = \frac{ze}{zf} = \ldots\ldots = \frac{xa+yc+ze+\ldots.}{xb+yd+zf+\ldots.}$

RAPID FIRE ROUND

$\frac{x}{y} = \frac{3}{2} \xrightarrow{\text{c+d}} \frac{x+y}{x-y} = \frac{3+2}{3-2} = \frac{5}{1} = 5$

(i) If $\frac{x}{3} = \frac{y}{2}$, then $\frac{x+y}{x-y}$ = 5 .

(ii) If $(4x - 3y) : (2x + 5y) :: 12 : 19$, then $x : y$ is _____.

$x : y \rightarrow \frac{x}{y}$

(iii) If $\frac{x+y}{2} = \frac{y+z}{3} = \frac{z+x}{4}$, then $x : y : z$ is _____.

$\frac{4x-3y}{2x+5y} = \frac{12}{19} \Rightarrow \frac{y\left(\frac{4x}{y} - 3\right)}{y\left(\frac{2x}{y} + 5\right)} = \frac{12}{19}$

Say $\frac{x}{y} = k$

$\frac{4k-3}{2k+5} = \frac{12}{19}$

$4 \times 19k - 3 \times 19 = 12 \times 2k + 12 \times 5$

$52k = 117$

$k = \frac{117}{52} = \frac{x}{y}$

RAPID FIRE ROUND

(i) If $\frac{x}{3} = \frac{y}{2}$, then $\frac{x+y}{x-y} =$ _____.

(ii) If $(4x - 3y) : (2x + 5y) :: 12 : 19$, then $x : y$ is _____.

(iii) If $\frac{x+y}{2} = \frac{y+z}{3} = \frac{z+x}{4}$, then $x : y : z$ is $\frac{3k}{2} : \frac{k}{2} : \frac{5k}{2} \Rightarrow 3 : 1 : 5$ (×2/k)

$\frac{x+y}{2} = \frac{y+z}{3} = \frac{z+x}{4} = k$

$x + y = 2k$
$y + z = 3k$
$z + x = 4k$
add
$(x+y+z) = \frac{9k}{2}$

$z = \frac{9k}{2} - 2k = \frac{5k}{2}$

$x = \frac{9k}{2} - 3k = \frac{3k}{2}$

$y = \frac{9k}{2} - 4k = \frac{k}{2}$

Module Question

Solve the equation $\frac{3x^4 + x^2 - 2x - 3}{3x^4 - x^2 + 2x + 3} = \frac{5x^4 + 2x^2 - 7x + 3}{5x^4 - 2x^2 + 7x - 3}$

C+d

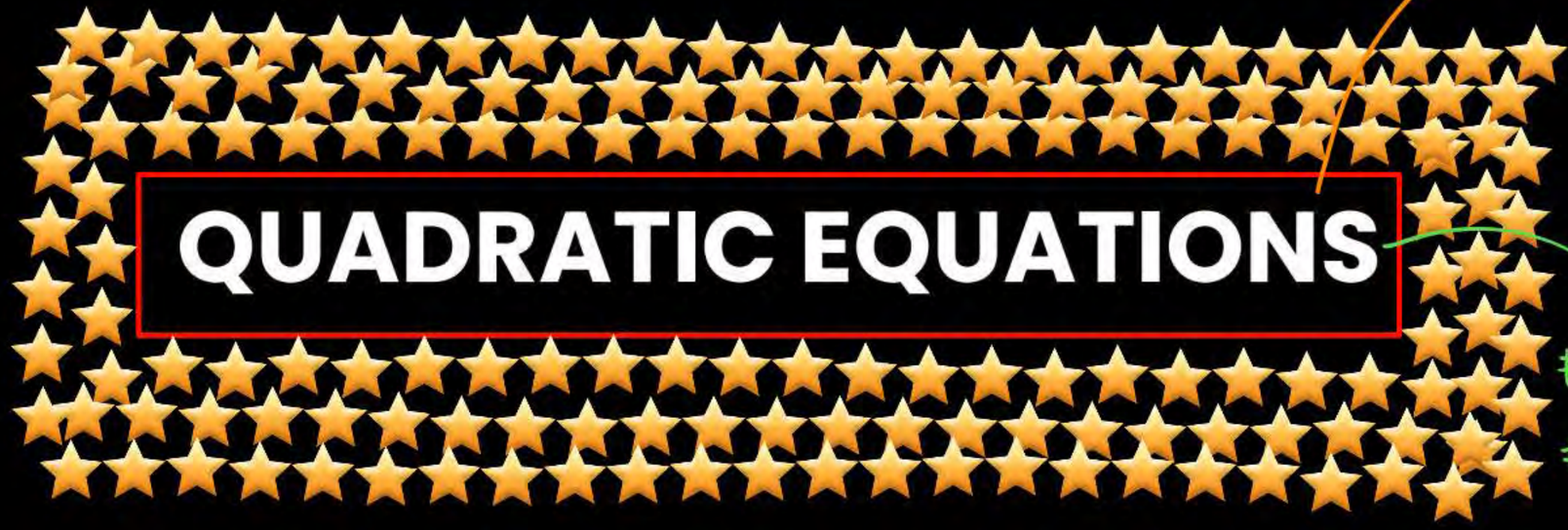


A $x = 5, 2$

B $x = 4, 1$

C $x = 3, 8$

D $x = 1, 5$



"easy"

Mains → 1 Ques^{n}.

Adv. → 1 Ques^{n}.

Used in almost every chapter.



POLYNOMIAL EQUATIONS

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots\ldots$$

Highest power of x present = degree.

Linear Equation: (degree = 1)

$$ax + b = 0,\ a \neq 0$$

Quadratic Equation: (degree = 2)

$$ax^2 + bx + c = 0,\ a \neq 0$$

Cubic Equation: (degree = 3)

$$ax^3 + bx^2 + cx + d = 0,\ a \neq 0$$

Quartic Equation: (degree = 4)

$$ax^4 + bx^3 + cx^2 + dx + e = 0,\ a \neq 0$$

Constant polynomial

$p(x) = k$, (degree = 0)



QUADRATIC EQUATION

$f(x) = x^2 + 3x - 7$ → Quad. expression

$x^2 + 3x - 7 = 0$ → Quad. eqⁿ.



QUADRATIC EQUATION

$$ax^2+bx+c=0\ ,\ (a\neq 0)$$

$$a\left(x^2+2\left(\frac{b}{2a}\right)x+\frac{c}{a}\right)=0$$

$$a\left(x^2+2\left(\frac{b}{2a}\right)x+\frac{b^2}{4a^2}-\frac{b^2}{4a^2}+\frac{c}{a}\right)=0$$

$$\left(x+\frac{b}{2a}\right)^2=\frac{b^2}{4a^2}-\frac{c}{a}$$

$$\left(x+\frac{b}{2a}\right)^2=\frac{b^2}{4a^2}-\frac{c}{a}$$

$$\left(x+\frac{b}{2a}\right)^2=\frac{b^2-4ac}{4a^2}$$

$$\left(x+\frac{b}{2a}\right)=\pm\sqrt{\frac{b^2-4ac}{4a^2}}$$

$$x=\frac{-b}{2a}\pm\frac{\sqrt{b^2-4ac}}{2a}$$

$$\boxed{x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}}$$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$ax^2 + bx + c = 0$

$$\alpha = \frac{-b + \sqrt{b^2-4ac}}{2a}$$

$$\beta = \frac{-b - \sqrt{b^2-4ac}}{2a}$$

degree = 2 = no. of solutions

$D = b^2 - 4ac$

Sum of Roots

$$\alpha + \beta = \frac{-b}{a}$$

Product of Roots

$$\alpha\beta = \frac{+c}{a}$$

Difference of Roots

$$|\alpha - \beta| = \left|\frac{\sqrt{D}}{a}\right| = \frac{|\sqrt{D}|}{|a|} = \frac{\sqrt{D}}{|a|}$$

Ques:-

$3x^2 + x - 7 = 0$ $\begin{cases} \alpha \\ \beta \end{cases}$

1. $\alpha + \beta = -\frac{b}{a} = \frac{-1}{3}$
2. $\alpha\beta = +\frac{c}{a} = +\frac{(-7)}{3} = -\frac{7}{3}$
3. $|\alpha - \beta| = \left|\frac{\sqrt{D}}{a}\right| = \left|\frac{\sqrt{85}}{3}\right| = \frac{\sqrt{85}}{3}$

$3x^2 + 1x - 7 = 0$

$a = 3$, $b = 1$, $c = -7$.

$D = b^2 - 4ac$

$= 1^2 - 4(3)(-7)$

$= 85$.

Question

If α & β are roots of the equation $ax^2 + bx + c = 0$ and $\beta = 2\alpha$ then correct relation between a, b & c is:

A $9b^2 = 2ac$

B $9b^2 = 4ac$

C $b^2 = 4ac$

D $2b^2 = 9ac$

$* \alpha + \beta = -\frac{b}{a}$, $* \alpha\beta = \frac{c}{a}$

$\alpha + 2\alpha = -\frac{b}{a}$

$3\alpha = -\frac{b}{a}$

$\alpha = -\frac{b}{3a}$

$\alpha(2\alpha) = \frac{c}{a}$

$2\alpha^2 = \frac{c}{a}$

$\alpha^2 = \frac{c}{2a}$

$\left(\frac{-b}{3a}\right)^2 = \frac{c}{2a} \Rightarrow \frac{b^2}{9a^2} = \frac{c}{2a} \Rightarrow 2b^2 = 9ac$

Question

If the equation $(k-2)x^2 - (k-4)x - 2 = 0$ has difference of roots as 3, then the value of k is

A. 1, 3

B. $3, \frac{3}{2}$

C. $2, \frac{3}{2}$

D. $\frac{3}{2}, 1$

$||x|| = |x|$

59%

$a = k-2$
$b = -(k-4)$
$c = -2$

$D = (-(k-4))^2 - 4(k-2)(-2)$

$= k^2 + 16 - 8k + 8(k-2)$

$= k^2 + 16 - 8k + 8k - 16$

$= k^2$

diff. of Root $= \left|\frac{\sqrt{D}}{a}\right| = 3$

$\left|\frac{\sqrt{k^2}}{k-2}\right| = 3$

$\frac{|k|}{|k-2|} = \frac{||k||}{|k-2|} = 3$

$|k| = 3|k-2|$

$k = 3(k-2)$ $\quad -k = 3(k-2)$

$k = 3k-6$ $\quad -k = 3k-6$

$6 = 2k$ $\quad 6 = 4k$

$3 = k$ $\quad \frac{3}{2} = k$

DO IT BY YOURSELF (DIBY)

QUESTIONS ON GRAPH

DIBY-107

If $|x-2|+2|x+2|=P$, then find P is given equation has

(i) no solution
(ii) 1 solution
(iii) 2 solutions
(iv) more than 2 solutions

Ans. :

(i) No solution $P<4$
(ii) 1 solution $P=4$
(iii) 2 solutions $P>4$
(iv) More then 2 solution $P\in\phi$

DIBY-108

Find the value of k if equation $|x+2|-2|x-7|+3|x-4|=k$ has

(i) one solution (ii) two solutions
(iii) 3 solutions (iv) 4 solutions
(v) no solution (vi) ∞ solutions

Ans. :

(i) one solution $k\in\phi$
(ii) 2 solutions $k>0$
(iii) 3 solutions $k\in\phi$
(iv) 4 solution $k\in\phi$
(v) no solution $k<0$
(vi) ∞ solutions $k=0$

QUESTIONS ON GRAPH

DIBY-109

Draw graph of $y = ||x-2|-3|$ and answer following questions:

(i) Minimum value of y is _____.

(ii) If $y = k$ has 2 solutions then k is _____.

(iii) If $y = k$ has 3 solutions then k is _____.

(iv) If $y = k$ has 4 solutions then k is _____.

(v) If $y = k$ has no solution then k is _____.

(vi) Number of solutions of : $||x-2|-3| = \frac{2x+10}{15}$, is(are) _____.

Ans. :

(i) 0

(ii) $k \in (3, \infty) \cup \{0\}$

(iii) $k = 3$

(iv) $k \in (0, 3)$

(v) $k \in (-\infty, 0)$

(vi) 4

QUESTIONS ON IRRATIONAL EQUATIONS & INEQUALITY

DIBY-110

It x is a real number, find the number of integral values of x for which the expression $\sqrt{|x-2|-3}+\sqrt{5-|x+1|}$ is defined.

[Ans. : 6]

DIBY-111

If the complete solution set of the inequality $1-\sqrt{1-4x^2}<3x$ is $(a,b]$, find the value of $10a+2b$.

[Ans. : 1]

DIBY-112

Solve : $\sqrt{x-2}\geq\sqrt{5-x}$

[Ans. : $x\in\left[\frac{7}{2},5\right]$]

DIBY-113 (Challenger)

Solve : $\sqrt{x-6}-\sqrt{10-x}\geq 1$

[Ans. : $x\in\left[8+\frac{\sqrt{7}}{2},10\right]$]

MISCELLANEOUS QUESTIONS

DIBY-114

Which of the following is true?

(A) $(\log_{10} 2)^2 + 1 > \log_{10} 4$

(B) $\log_{10} 90 > \log_5 50$

(C) $\log_4 \log_3 \log_2 16 > \log_{16} 4$

(D) $2(\log_{10} 3)^2 - 3(\log_{10} 2)^2 > (\log_{10} 2) \times (\log_{10} 3)$

[Ans. : A, D]

DIBY-115 (Challenger)

Positive numbers x, y **and** z **satisfy** $xyz = 10^{81}$ **and** $(\log_{10} x)(\log_{10} yz) + (\log_{10} y)(\log_{10} z) = 468.$

Find the value of $(\log_{10} x)^2 + (\log_{10} y)^2 + (\log_{10} z)^2$

[Ans. : 5625]

DIBY-116

Consider the system of equations $\log_{10}(2000xy) - \log_{10} x \cdot \log_{10} y = 4$, $\log_{10}(2yz) - \log_{10} y \cdot \log_{10} z = 1$ **and** $\log_{10}(zx) - \log_{10} z \cdot \log_{10} x = 0$, $x, y, z \leq 90$

then $x + y + z =$

[Ans. : 7]

DIBY-117

Let $S = \left\{x \in R: \left(\sqrt{3}+\sqrt{2}\right)^x + \left(\sqrt{3}-\sqrt{2}\right)^x = 10\right\}$. Then the number of elements in S is

[JEE Mains-2024]

(A) 4 (B) 0 (C) 2 (D) 1

[Ans. : (C)]

DIBY-118

The number of elements in the set $\left\{n \in \mathbb{Z}: \left|n^2 - 10n + 19\right| < 6\right\}$ is

[JEE Mains-2023]

[Ans. : 6]

DIBY-119

Let $S = \left\{x \in [-6,3] - \{-2,2\}: \frac{|x+3|-1}{|x|-2} \geq 0\right\}$ and $T = \left\{x \in Z: x^2 - 7|x| + 9 \leq 0\right\}$. Then the number of elements of $S \cap T$ is

[JEE Mains-2022]

(A) 7 (B) 5 (C) 4 (D) 3

[Ans. : (D)]

MISCELLANEOUS QUESTIONS

DIBY-120

Let $A = \{x: |x^2 - 10| \leq 6\}$ and $B = \{x: |x - 2| > 1\}$. Then [JEE Mains-2026]

(A) $A - B = [2, 3)$

(B) $A \cup B = (-\infty, 1] \cup (2, \infty)$

(C) $B - A = (-\infty, -4) \cup (-2, 1) \cup (4, \infty)$

(D) $A \cap B = [-4, -2] \cup [3, 4]$

[Ans. : (C)]

DIBY-121

If $A = \{x \in R: |x - 2| > 1\}, B = \{x \in R: \sqrt{x^2 - 3} > 1\}, C = \{x \in R: |x - 4| \geq 2\}$ and Z is the set of all integers, then the number of subsets of the set $(A \cap B \cap C)^C \cap Z$ is [JEE Mains-2021]

[Ans. : 256]

DIBY-122

The number of elements in the set $\{x \in R: (|x| - 3)|x + 4| = 6\}$ is equal to [JEE Mains-2021]

(A) 3

(B) 2

(C) 4

(D) 1

[Ans. : (B)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: Basic Maths (4 Sittings Each of 1 Hr.)

Exercise-03(BRIDGE): COMPLETE

Exercise-04(JEE ADV. LEVEL):COMPLETE

} 4HR.

It is never about end result

It's always about Journey

Thank You

#futureIITians